compute vector-Jacobian products on-the-fly

works with any parameterization

leveraging existing, memory-efficient ops like log-sum-exp

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#3D251F cocap	#3DA2B7 teal	#ED6758 salmon	#FEC44C citrus	#D9B37E beige	#AA664A copper

Autodiff: advantages

- **Avoids Explicit Jacobian Computation**: Autodiff frameworks usually **compute vector-Jacobian products on-the-fly**, avoiding full Jacobian construction.
- **GPU-Accelerated and ML—Friendly**: Autodiff integrates seamlessly with modern libraries (e.g., PyTorch, JAX), enabling efficient batched computation and backpropagation on GPU.
- **Handles arbitrary models**: **works with any parameterization** of $a(\theta)$, $x(\theta)$, or $c_\phi(x, y)$, including deep networks, regardless of complexity.
- **Exploits optimized ops**: Autodiff traces gradients through full Sinkhorn loop, **leveraging existing, memory-efficient ops like log-sum-exp** already built into DL libraries.
- **Flexible Accuracy-Speed Tradeoff**: Tune number of Sinkhorn iterations or use .detach()-based approximations to balance gradient fidelity and computational efficiency.

Stop Grad Trick